Setting up the Lab Environment

6COSC019W

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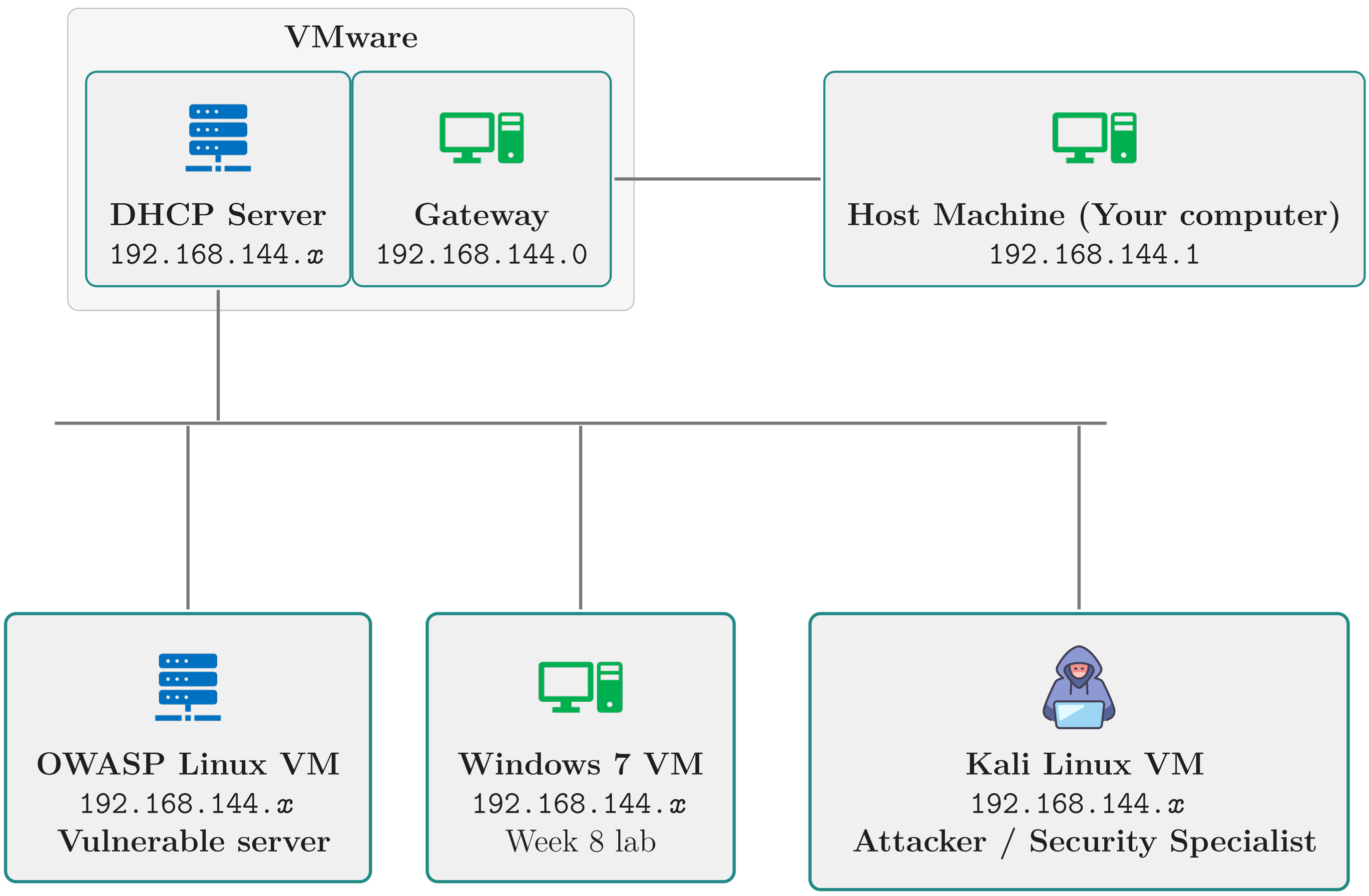
Lab environment



Please read this before you start

- This document explains how to download and set up the virtual machines (VMs). In most labs, the main machines are already available in `C:\VirtualMachines` ; you only need to add them to VMware.
- Before downloading a VM, **it is essential to first check** `C:\VirtualMachines` for the relevant `.7z` archives. Most labs already provide the `.7z` files there.
- Always download from the correct local server; using the wrong server will be significantly slower.
- Only download to `C:\VirtualMachines` . If you downloaded elsewhere, move the files into that folder to avoid filling the system drive.
- Download one VM at a time. They are large files; parallel downloads can cause slowdowns or failures.
- Shut down the VM before logging out; otherwise you may not be able to use it next time.
- If a VM becomes unusable, remove it from VMware (including deleting the VM's files) and re-import it from the original archives.
- **Do not delete the `.7z` files.** They are the source for restoring or re-importing VMs quickly.

1 Virtual Machines and Download locations

- During the course, several VMs are used. These can be obtained from the Virtual Machines page.

- Use the correct download location depending on where you are:

  - ↳ On campus (Eduroam or any lab with VMware Workstation, except CLG.43 and CLG.45): http://10.20.144.78/download/VMs/
  - ↳ In CLG.43 (Network Lab): http://192.168.143.203/download
  - ↳ In CLG.45 (Network Lab): http://192.168.145.206/download
  - ↳ Outside the university:
  - ↳ 1) Connect to the VPN: https://remote2.westminster.ac.uk/
  - ↳ 2) VPN setup guidance: https://support.cse.westminster.ac.uk/w/index.php/VPN
  - ↳ 3) When prompted for the portal address, use `remote2.westminster.ac.uk`
  - ↳ 4) After connecting, access: https://download.cse.westminster.ac.uk/VirtualMachines/ and download the required VM

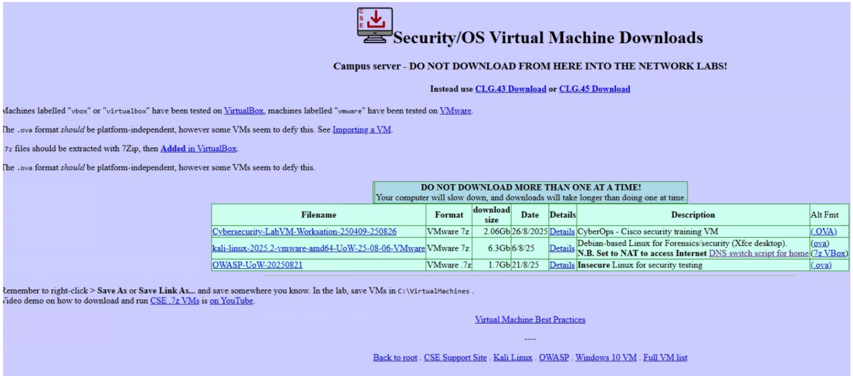


Figure 1.1: The virtual machines images download page

- The Virtual Machines used in this module are:
  - ↳ Security Analyst/Attacker: `kali-linux-2025.2-vmware-amd64-UoW-25-08-06-VMware`
  - ↳ Linux vulnerable machine (OWASP): `OWASP-UoW-20250821`
  - ↳ Windows Client machine: `Windows-7-IE-25-06-20`

VM downloads

> Only download VMs needed for your module.

- The username and password for each VM are listed below.
- Kali Linux credentials:
  - ↳ username: `kali`
  - ↳ password: `kali`
- OWASP BWA VM (Linux) credentials:
  - ↳ username: `root`
  - ↳ password: `owaspbwa`
- Windows client login download information and login details will be provided when required.

2 Install software on personal devices

2.1 7-zip for both Windows and macOS

- Download and install 7-Zip to extract the module virtual machine files.
  - ↳ 7-Zip is required to unpack large compressed archives such as `.7z` or `.zip` files.

- ↳ 7-Zip is available for both Windows and macOS.
- ↳ Download link: https://www.7-zip.org/download.html

2.2 Install VMware Workstation on Windows

- Download VMware Workstation (Windows) or VMware Fusion (macOS) from `https://www.vmware.com/products/desktop-hypervisor/workstation-and-fusion`.
  - ↳ You may need to register an account.
  - ↳ As of late 2024, VMware Workstation Pro and VMware Fusion are free for personal, educational, and commercial use (no licence key required).

2.3 Install VMware Fusion On macOS

- Download VMware Fusion from `https://www.VMware.com/products/desktop-hypervisor/workstation-and-fusion`.
  - ↳ Use VMware Fusion with caution on macOS, as some versions may be unstable or have limited support for certain virtual machines, especially on Apple Silicon devices.

Apple Silicon (M1/M2/M3) Compatibility Note

- Macs with Apple Silicon processors use the ARM64 architecture.
- Many Cyber Security virtual machines are built for x86_64 (Intel) systems and may not run correctly on Apple Silicon.
- Some VMs may fail to boot, run slowly, or have missing network or kernel features.
- Where possible, use university lab machines or Windows PCs for full compatibility.
- VMware Fusion on Apple Silicon should be used with caution for this reason.
- This is a limitation not only for VMware but for all virtualisation software on Apple Silicon devices.

3 Working on University Machines

- On university machines, VMware Workstation is installed in CLG.43, CLG.45 and will also be available in CLG.42, LG.105, 2.110 and 2.112. Guidance: VMware support page.
- For most lab activities, we will be using two module virtual machines.
- **Kali Linux VM**
  - ↳ This is the attacker or penetration testing machine with pre-installed security tools.
  - ↳ The university Kali VM is customised and different from the public Kali version.
  - ↳ Latest university version: **kali-linux-2025.2-vmware-amd64-UoW-25-08-06-VMware.7z**

- **OWASP Broken Web Apps VM**
  - ↳ This is the vulnerable machine containing multiple services such as web applications and databases.
  - ↳ Latest university version: **OWASP-UoW-20250821.7z**

3.1 Only if files are not in the C:\VirtualMachines folder

- The virtual machines should already be available in the **C:\VirtualMachines** folder.
  - ↳ If they are present, go to Section 4.
- If the `.7z` files are not available, download them from the appropriate link below.
- **For CLG.43**
  - ↳ http://192.168.143.203/download
- **For CLG.45**
  - ↳ http://192.168.145.206/download
- **For internal, eduroam or CLG.42**
  - ↳ http://10.20.144.78/download/VMs/
- Download the required VMs and save them in the **C:\VirtualMachines** folder.

3.2 Working on University Machines Remotely

- Splashtop gives you remote access when you are unable to use your personal device and are working from outside the university campus.
  - ↳ It allows you to connect to university lab machines and run VMware remotely.
- Splashtop is installed in CLG.42 and LG.105, allowing you to access and run VMware remotely from your own device.
  - ↳ Sign in to Splashtop at https://my.splashtop.com/login/sso.
  - ↳ Search for machines that start with either `CLG.42` or `LG.105`. See Figure 3.1.
  - ↳ You can also download the Splashtop application for easier access. Click on **Business App** to download it, as shown in Figure 3.1.

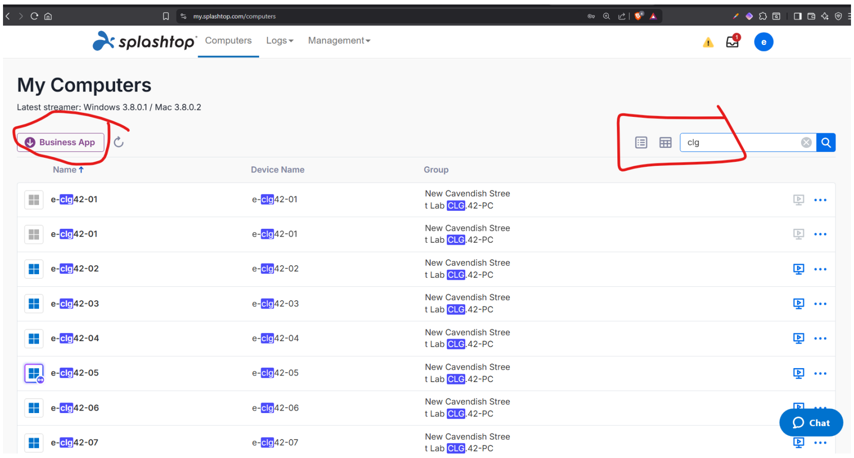


Figure 3.1: Splashtop page

4 Unpacking the VMs

- By now, you should have the two virtual machines downloaded.
- On university machines, the VMs should already be located in **C:\VirtualMachines** as shown in Figure 4.1.
- You should have the following machines:
  - ↳ [1] Kali Linux VM
  - ↳ [2] OWASP Broken Web Apps VM

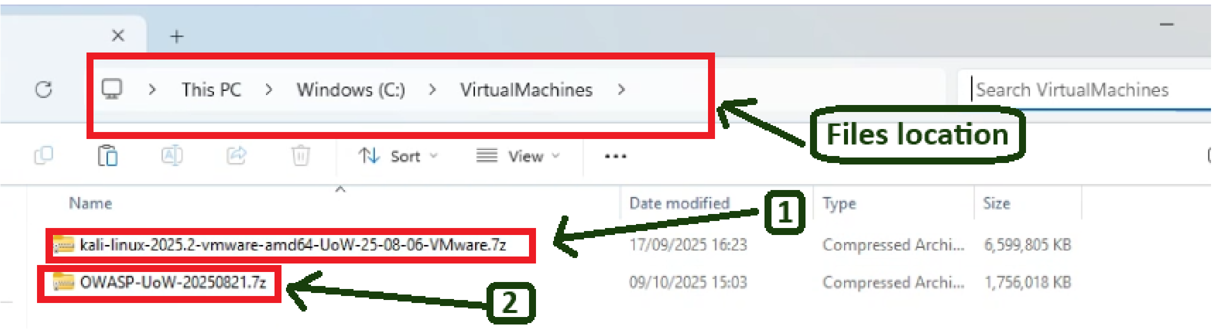


Figure 4.1: Downloaded VMs

- We can extract the 7-zip files in any location where we need our Virtual Machines.
- If you are on a university machine, this would be in the same folder **C:\VirtualMachines** See Figure 4.2.
  - ↳ [1]- Right-click on any of the 7-zip files
  - ↳ [2]- select **7-zip**
  - ↳ If you don't see 7-zip. select first "**show more options**"

↳ [3]- Select **Extract Here**

➔ Repeat these steps for the second VM file.

➔ If you are doing this on your own machine, you can set them up in the folder you will be using for your Virtual Machines. that can be in any drive that you have space.

↳ In this case, you select **Extract this** to choose the location to extract the file to.

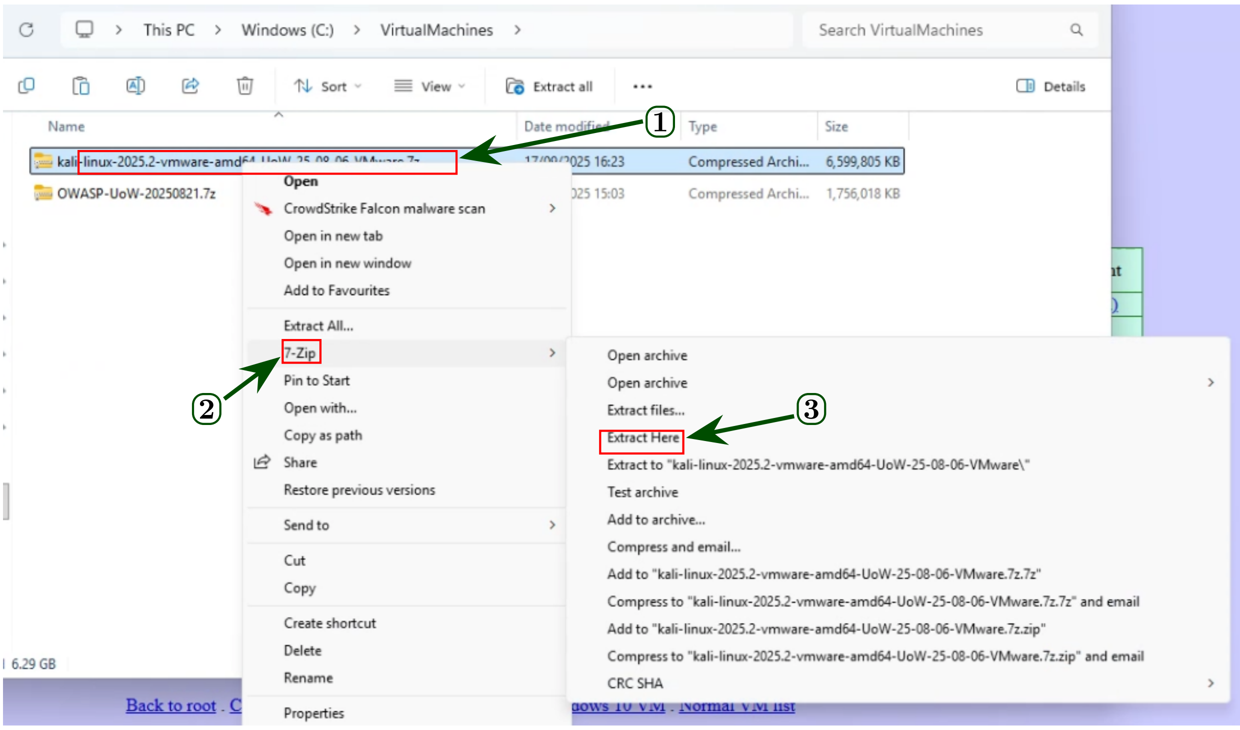


Figure 4.2: Extract VMs

➔ You should now have two files and two folders.

- At this stage, you should end up with the two 7z files and the two folders of the extracted virtual machines (Fig.4.3)

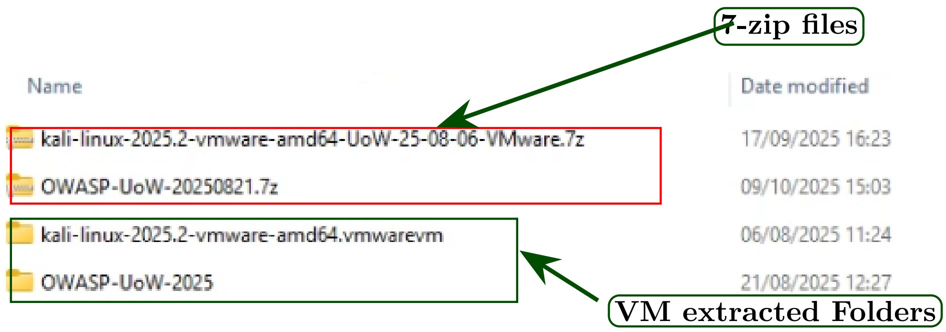


Figure 4.3: VirtualMachines folder contents

5 Deploying Virtual Machines on VMware

VMware: Open vs Import

Most Cyber Security VMs are provided as ready-built VMware folders (**.vmx** + **.vmdk**); importing **.ova** is rarely needed.

We start by deploying the first VM. In my example. I am first deploying the Kali Linux VM. We will need to do this for both VMs and any other VMs we will need later on..

We will first start VMware that is already pre-installed on the machine. In VMware Workstation:

↳ Go to **File** → **Open** and select the **.vmx** file. The VM is added directly to your library.

We can also use the Open A virtual Machine icon in VMware Workstation following the steps below (shown in Figure5.1):

↳ [1]- Click on **Open a Virtual Machine**

↳ [2]- Navigate to the folder of each VM you extracted

↳ [3]- Select the VM vmx file

↳ [4]- Click Open

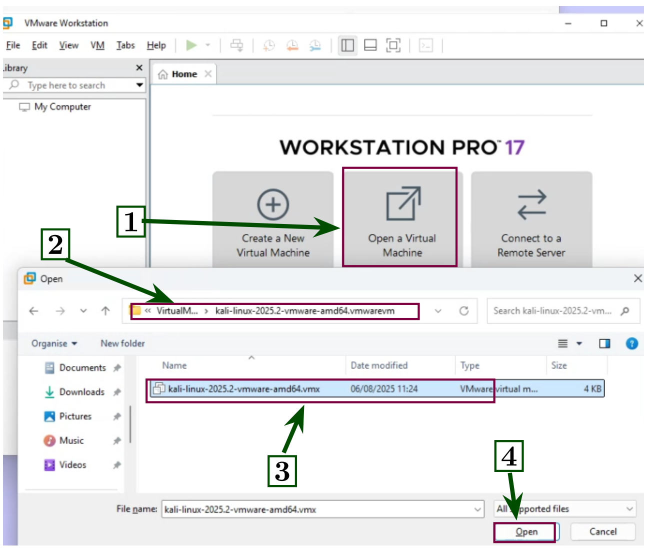


Figure 5.1: Deploy Virtual Machine

We repeat the steps above for any Virtual Machine we need to deploy on VMware.

↳ [1] We now have both VMs deployed as shown in Figure5.2.

↳ [2] When we click on one of them, it will open a window tab within the VMware

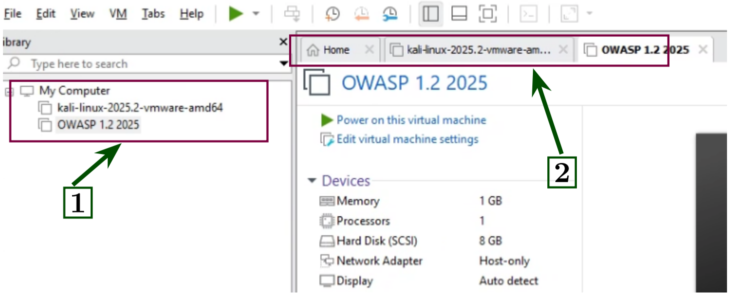


Figure 5.2: Kali and OWASP Deployed on VMware

5.1 Removing deployed Virtual Machines on VMware

Starting VMware from a clean state

- When using a university machine, it is best to start from a clean virtual machine state.
- If virtual machines are already deployed, you should remove them before starting.

- To remove a virtual machine:

  ↳ Right-click on the virtual machine you want to remove and select **Remove**.

  ↳ Confirm by selecting **Remove**.

  ↳ In the folder where your virtual machines are stored, delete the extracted folder of the virtual machine you just removed from VMware.

Do not delete the 7z file. You will need it to deploy a clean virtual machine again.

6 Network Setup

6.1 Virtual Network Editor - on your own machine

On a university machine

- Creating or modifying **virtual networks** in the VMware **Virtual Network Editor** requires **administrator access** and is only possible on your **personal machine**.
- On university lab machines, host-only networks are already pre-configured. Do **not** attempt to change them.

Network Configuration Note

Most labs require the **isolated host-only network** unless the instructions specify otherwise.

If you are using your own machine, follow these steps to **create a host-only virtual network**:

↳ Open **VMware Workstation** → **Edit** → **Virtual Network Editor** (See Figure 6.1).

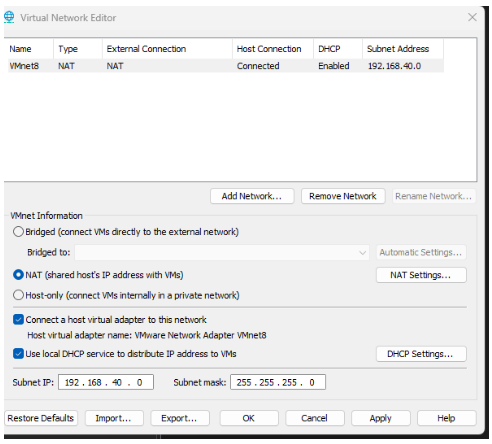


Figure 6.1: Virtual Network Editor: **Edit** → **Virtual Network Editor**

If this is the first time you set up the Virtual Network, you should only have a NAT interface. We will need to add the host-only interface, so that we have an isolated network.

↳ [1] Click **Change Settings (admin rights required)** (See Figure 6.2).

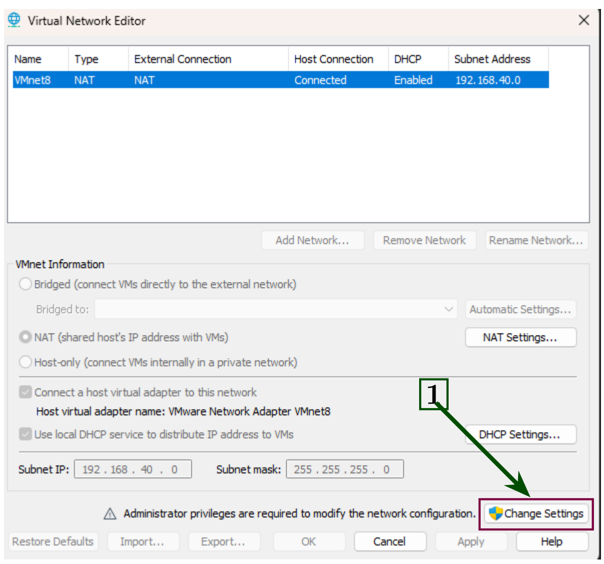


Figure 6.2: Change settings (needs administrator privilege)

↳ [1] Click on **Add Network** (See Figure 6.3).

↳ [2] choose a network interface name and click **OK** (e.g., VMnet0) (See Figure 6.3)

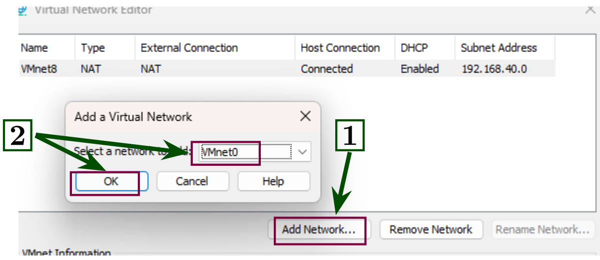


Figure 6.3: Add Network

For consistency with university lab setups, you should also change the Network settings and the DHCP settings to be the same as the university machines. This will also make it easier to follow the lab instructions. (See Figure 6.3)

↳ [1]- Select the Virtual machine network you have just created (in my case, it is VMnet0)

↳ [2]- Change the Subnet IP address to **192.168.144.0**

↳ [3]- Make sure it is on **Host-Only**. If not, change it to **Host-Only (Connect to a private network)**

↳ [4]- Press **Apply**

↳ [5] Enable the DHCP by selecting (**Use local DHCP to distribute IP addresses to VMs**).

↳ [6]- Click on **DHCP Settings**

↳ [7]- Make sure the starting IP and the ending IP addresses are set: starting IP `192.168.144.128` and the ending IP `192.168.144.254`. Press **Ok**

↳ [8] Press **OK** to close the network settings windows

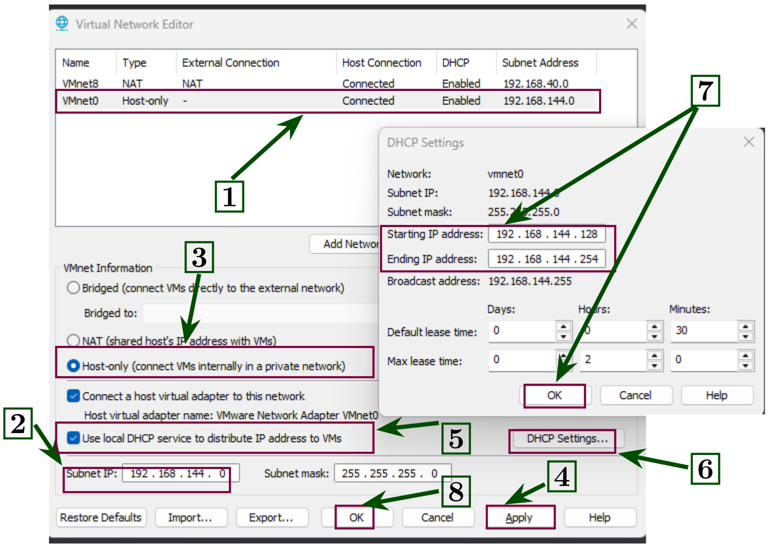


Figure 6.4: Network settings

6.2 Host-only Connected VM – Isolated Network

- After we have created the **host-only** network on your machine (pre-configured on university lab machines), we can now change the VM settings to connect it to either NAT (for Internet access) or Host-Only for an isolated private network.

- [1]Select the VM, then go to **VM** → **Settings**.
  - ↳ [2]Shortcut: Press `Ctrl+D`, or right-click the VM and choose **Settings**.
  - ↳ [3]Select **Network Adapter**
  - ↳ [4] For isolated private network, select **Host-only**.
  - ↳ [6] Press **OK** to close the settings.

This step does **not** require administrator privileges and must be done by **everyone** , including students using university lab PCs.

6.3 Internet Connected VM

Internet Connected Network

- Sometimes only the Kali VM needs Internet access.
- Do not connect the Windows 7, or the OWASP VMs to the Internet.

- To temporarily switch a VM's network to access the Internet via the host:
  - ↳ [5]- **Select NAT** to allow the VM to access the Internet through the host's connection.

↳ [6] Press **OK** to close the settings.

This step does **not** require administrator privileges and must be done by **everyone** , including students using university lab PCs.

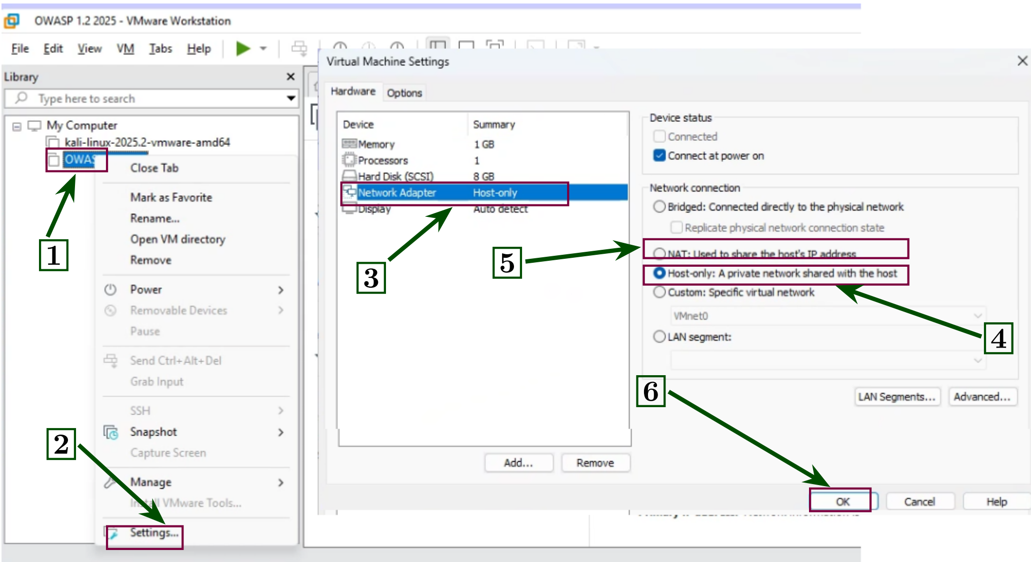


Figure 6.5: Host-only Adapter Settings or NAT

7 Starting a VM

Now we have the network settings ready, we can start each of the VMs.

↳ [1] Select the VM you want to start

↳ [2] Start the VM from the VMware interface using the **Start** button or by double-clicking the VM in the library.

↳ [2] (alternative) You can also start the VM from the menu. Select the VM from the top menu then go to **VM** → **Power** → **Start Up Guest**.

↳ [2] Shortcut: Press `Ctrl+B` to start the VM

↳ [3] The first time the virtual machine runs, a window will pop up asking whether the machine was moved or copied. You must select **I copied it**.

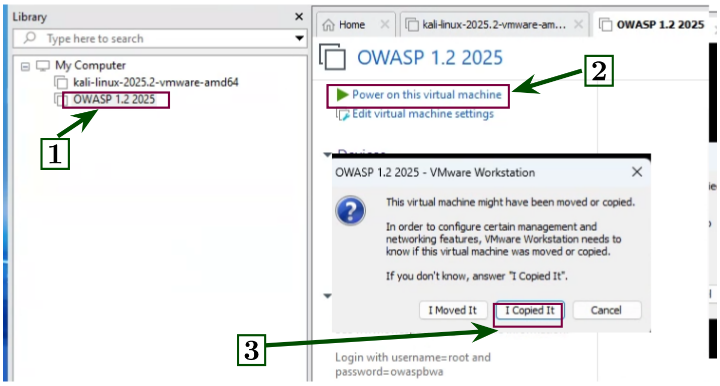


Figure 7.1: Host-only Adapter Settings or NAT

7.1 Opening a Terminal

- Most lab tasks require using the Linux terminal on Kali, and sometimes on the OWASP or Windows VM.
- To open a terminal in Kali:
  - ↳ Click the terminal icon in the top panel, or
  - ↳ Right-click the desktop and select **Open Terminal**.
  - ↳ You can also use the keyboard shortcut:

```
Ctrl + Alt + T
```

- The terminal is used to run commands, test networks, and use security tools.
- A detailed introduction is provided in the **Introduction to Linux** lab.

8 Network connectivity

8.1 Using NAT outside the university (DNS and resolvconf)

Internet connection outside the university (NAT)

- Some activities require only the Kali VM to be online (NAT).
- The university Kali image is configured for use inside the university network to avoid blocked Kali domains when updating the repository.

- For this reason, you need to switch the DNS resolver to your home network when working outside the university premises.
- The script can be downloaded from `github.com/6COSC019W` or cloned directly:

```
git clone https://github.com/azelhajjar/6COSC019W.git
```

- Navigate to the repository folder.
- In my case, it is on the Desktop.

```
cd ~/Desktop/6COSC019W
```

- First, make the script executable:

```
chmod +x switch-dns.sh
```

- Usage examples:

```
sudo ./switch-dns.sh uni # switch to University DNS
```

```
sudo ./switch-dns.sh home # switch to Home DNS
```

```
sudo ./switch-dns.sh status # check active profile
```

8.2 Identify your IP address

- To check your IP address on Linux (Kali):

```
ip a
```

  - ↳ The output may be cumbersome, as many network interfaces are listed.
  - ↳ Legacy alternative:

```
ifconfig eth0
```

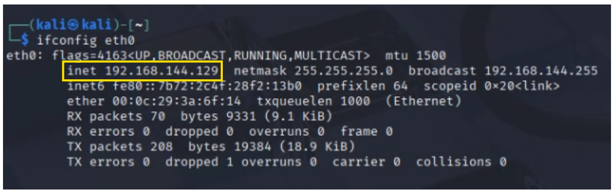


Figure 8.1: Kali IP address

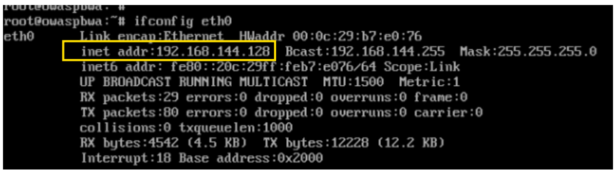


Figure 8.2: OWASP IP address

- To display the IP address for the host-only interface:

```
ip -4 a show eth0
```

! Some interfaces may show two IP addresses. This is normal and does not affect your lab work.

↳ Either IP address will behave the same for the lab activities.

↳ Alternatively, you can use:

```
ifconfig eth0
```

↳ The Kali IP on the host-only network is 192.168.144.129 (Fig.8.1).

↳ The OWASP IP on the host-only network is 192.168.144.128 (Fig.8.2).

Your IP address may be different, as it is assigned automatically using DHCP.

- Always check your own IP address as shown above.
- To check your IP address on Windows, open **cmd** and run:

```
ipconfig
```

- To test connectivity from Kali to the OWASP machine:

```
ping 192.168.144.128
```

- A host-only network isolates your VMs from the Internet while allowing internal communication.
- Devices must be on the same subnet to communicate.

8.3 Switching network settings

Resetting Kali's Network After Switching NAT←→ Host-only

- When switching between **NAT** and **Host-only**, the IP may not update automatically. Reset the interface:

```
sudo ifconfig eth0 down
sudo ifconfig eth0 up
sudo dhcpcd eth0
```

- Then confirm the new IP address:

```
ifconfig eth0
```

9 User Terminal

Very important for the lab assignment

- When working in the lab on Kali, you will take screenshots for your assignment.
- Screenshots without your student ID will receive a zero mark.
- Run the **student ID script** at the start of every lab session.

- Open a terminal and type:

```
./studentid.sh
```

  ↳ This will add your student ID.

  ↳ The script is in the home folder (/home/kali/). if you are somewhere else, you first need to go to the folder.

```
cd /home/kali/
```

- You will get a prompt to enter your student ID (Fig.9.1).

  ↳ Once you enter the ID, press enter.

  ↳ Close the terminal for the changes to apply.

- The terminal now has your ID next to the user logged in (Fig.9.2).

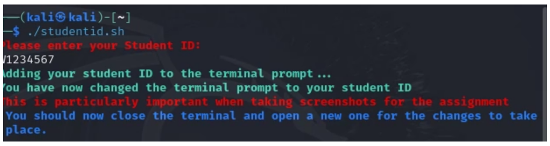


Figure 9.1: Execute the script and enter your User ID prompt

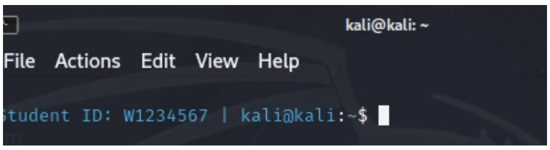


Figure 9.2: New User ID terminal prompt. For each session, you need to either create a new kali VM or setup a new one!

10 Problems that you can possibly face

- Ensure your VM's virtual network is connected in VMware.
- On Kali, network changes may require restarting NetworkManager:

```
sudo service network-manager restart
```

- If the machines fail to start, the error usually indicates:
  - Insufficient disk space. Free up space.
  - Virtualisation not enabled in BIOS. See the Microsoft guide: link
- If you cannot see the Host-Only adapter, the network was not created (Fig.6.3).
  - If the IP does not change, run **sudo dhcpcd**.
- No Internet on NAT. See Section 8.1.